

Introduction to HTML

HTML (HyperText Markup Language) is the standard markup language used to create web pages.

- HTML describes the structure and content of a webpage.

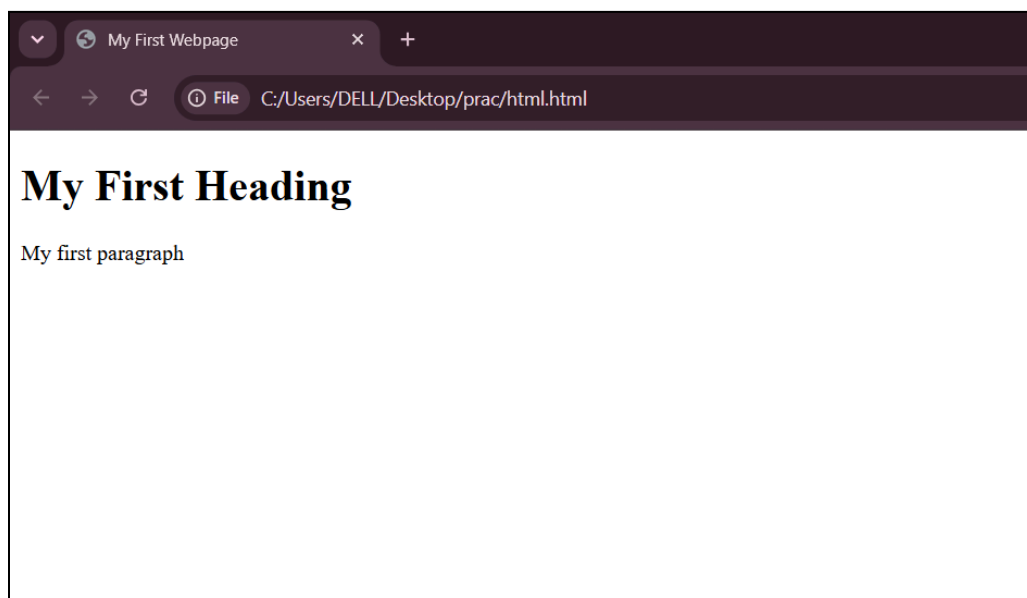
Example:

```
<!DOCTYPE html>
<html>

<head>
<title>My First Webpage</title>
</head>

<body>
<h1>My First Heading</h1>
<p>My first paragraph</p>
</body>

</html>
```



Explanation:

Tag	
<code><!DOCTYPE html></code>	defines this is an HTML5 document
<code><html></code>	Root element
<code><head></code>	Metadata about the webpage
<code><body></code>	Visible contents
<code><title></code>	specifies a title for the webpage, shown in the title bar
<code><h1></code>	defines a large heading
<code><p></code>	defines a paragraph

- ☐ Metadata means data about data. It refers to hidden information about a web page that describes its content, structure, and behavior to search engines, browsers, and other web services.

HTML Elements

- HTML consists of a series of elements
- HTML elements tell the browser how to display the content

An HTML element is defined by a **start tag**, **some content**, and an **end tag**:

`<tagname>` Content `</tagname>`

Example:

`<p>` Hello World `</p>`

- Opening tag = `<p>`
- Content = Hello World
- Closing tag = `</p>`

The entire thing is called an **element**.

Nested Elements

Elements can be placed inside other elements:

```
<body>
  <p>Hello</p>
</body>
```

Parent, Child, and Sibling:

```
<body>

  <h1>Title</h1>

  <p>Paragraph 1</p>

  <p>Paragraph 2</p>

</body>
```

The structure is:

```
body
|__h1
|__p
|__p
```

Parent Element

Parents contain other elements.

Parent: `<body>`

Child: `<h1>`, `<p>`

Child Element

Elements inside another element.

`<h1>`, `<p>` are children of `<body>`.

Sibling Element

Elements having the same parent.

`<h1>`, `<p>`, `<p>` are siblings.

Block and Inline Elements

Every HTML element has a default display value, depending on what type of element it is. The two most common display values are block and inline.

1. Block Elements

Block-level elements always start on a new line, and the browsers automatically add some space before and after the element.

A block-level element always takes up the full width available (stretches out to the left and right as far as it can).

Two commonly used block elements are: `<p>` and `<div>`.

- The `<p>` element defines a paragraph in an HTML document.
- The `<div>` element defines a division or a section in an HTML document.

Examples:

```
<div>
<p>
<h1>
<hr>
<form>
<li>
<table>
```

div:

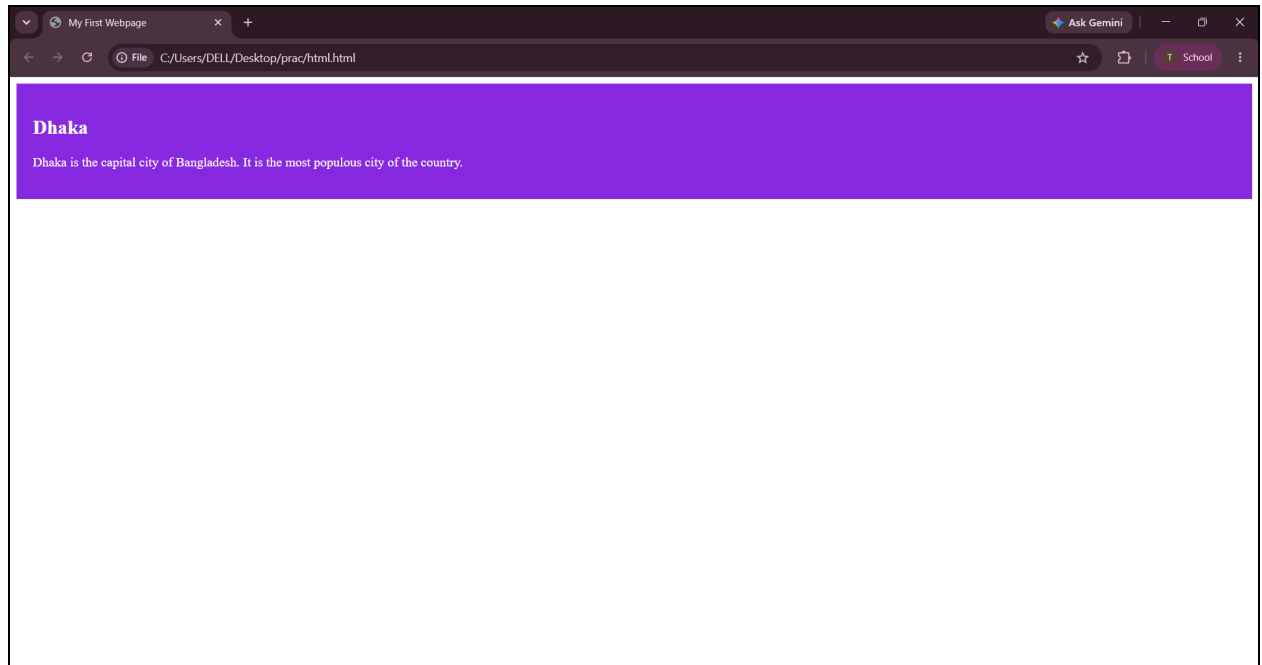
```
<!DOCTYPE html>
<html>

<head>
<title>My First Webpage</title>
</head>

<body>

<div style="background-color:blueviolet;color:white;padding:20px;">
  <h2>Dhaka</h2>
  <p>Dhaka is the capital city of Bangladesh. It is the most
  populous city of the country.</p>
</div>

</body>
</html>
```



2. Inline Elements

Inline elements stay in the same line. It only takes up as much width as necessary.

Examples:

`<span>`

`<b>`

`<i>`

`<a>`

`<br>`

`<sup>`

`<sub>`

span:

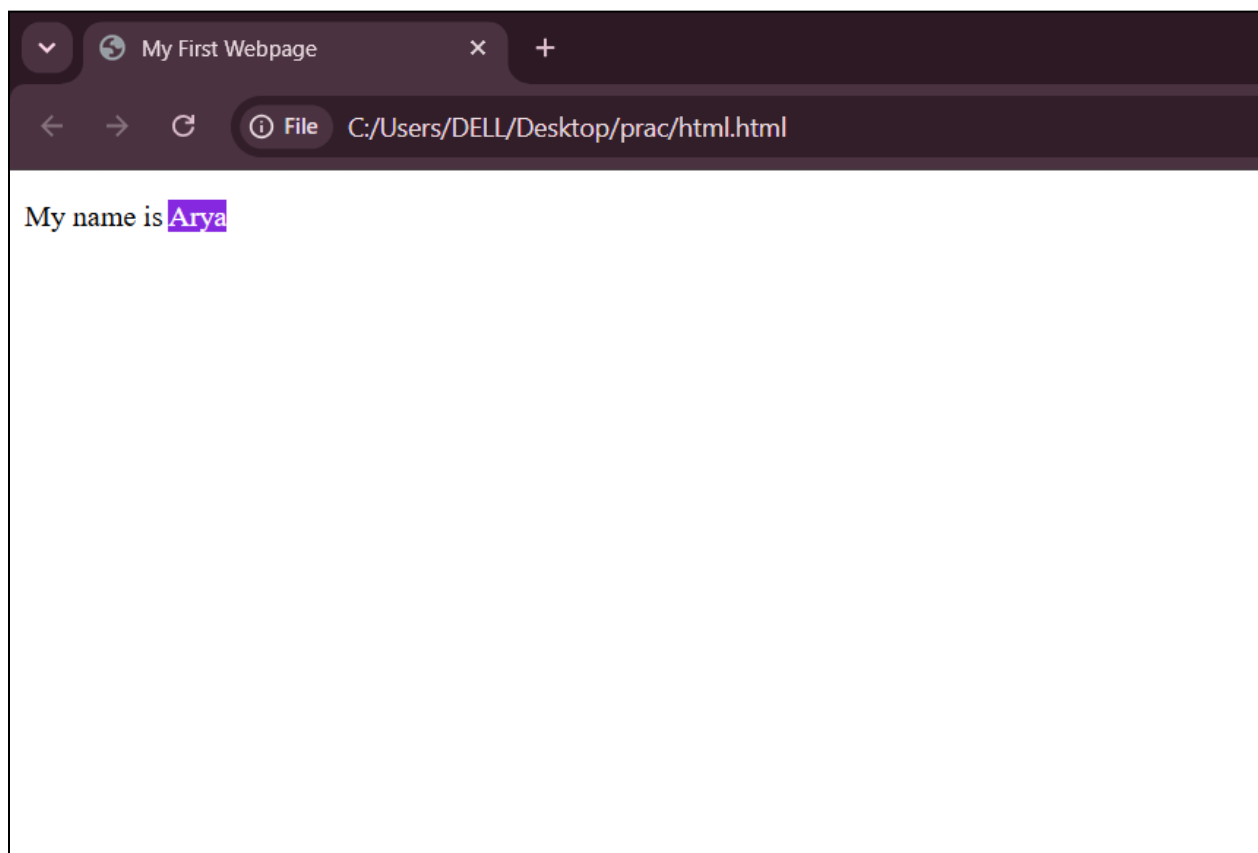
```
<!DOCTYPE html>
<html>

<head>
<title>My First Webpage</title>
</head>

<body>

<p>
    My name is
    <span
style="background-color:blueviolet;color:white">Arya</span>
</p>

</body>
</html>
```



HTML Headings

<body>

<h1>Heading 1</h1>

<h2>Heading 2</h2>

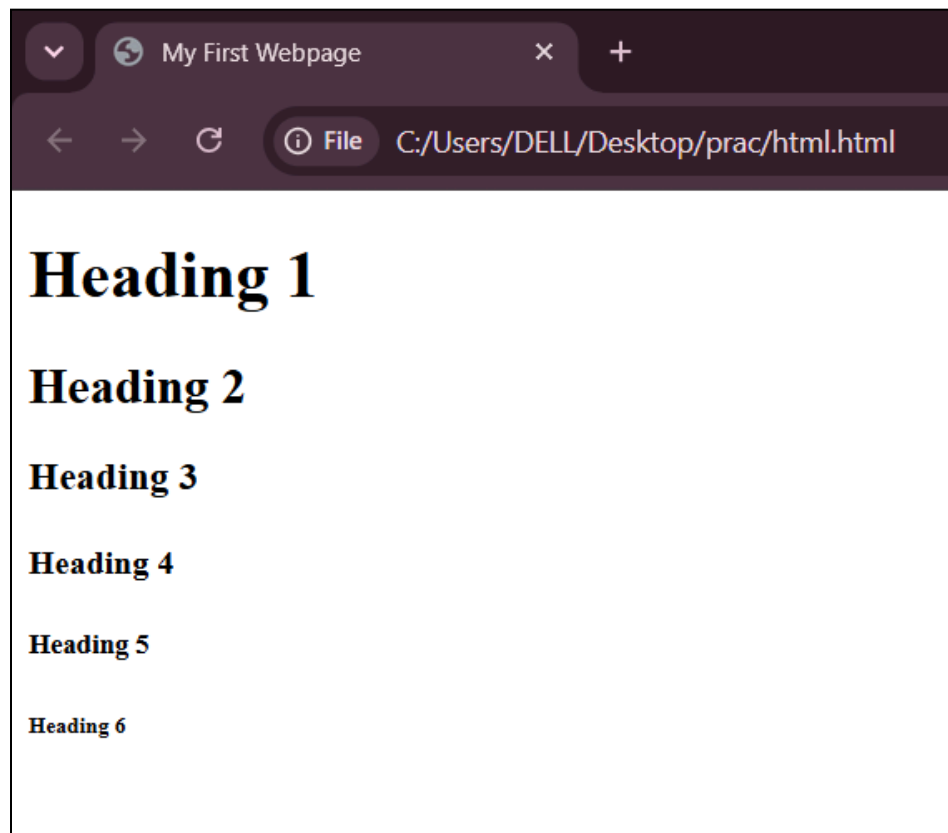
<h3>Heading 3</h3>

<h4>Heading 4</h4>

<h5>Heading 5</h5>

<h6>Heading 6</h6>

</body>

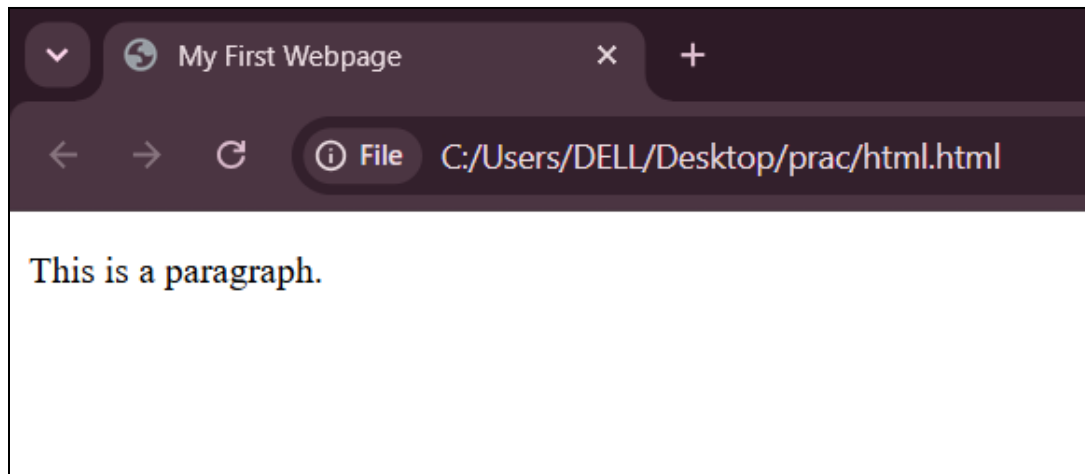


HTML Paragraphs

<body>

<p>This is a paragraph.</p>

</body>



Bold	Bold Text
Italic	<i>Italic Text</i>
Superscript	x²
Subscript	H₂O
Code	<code> int a = 10; </code>
Horizontal line (empty tag)	<hr>
Line break (empty tag)	

Example:

```
<body>
```

```
<p>
```

This is a paragraph.

```
<hr>
```

Bold Text *Italic Text* x² H₂O

```
<br>
```

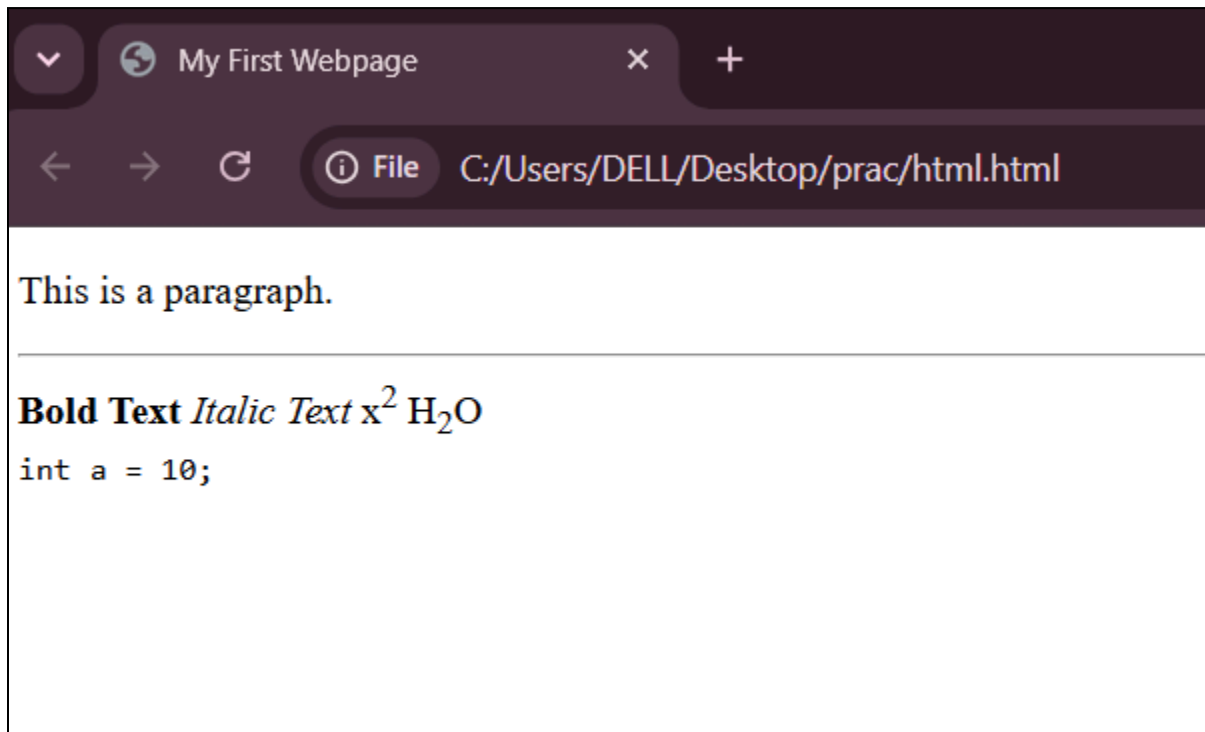
```
<code>
```

```
int a = 10;
```

```
</code>
```

```
</p>
```

```
</body>
```



HTML Comments

`<!-- This is a comment -->`

Example:

`<!-- Main Heading -->`

`<h1>Welcome</h1>`

HTML Attributes

HTML attributes provide additional information about HTML elements. Attributes are always specified in **the start tag**. Attributes usually come in name/value pairs like: **name = "value"**.

Example:

1. ``

Attribute name: src

Attribute value: cat.jpg

2. `<p style="color:red;">`

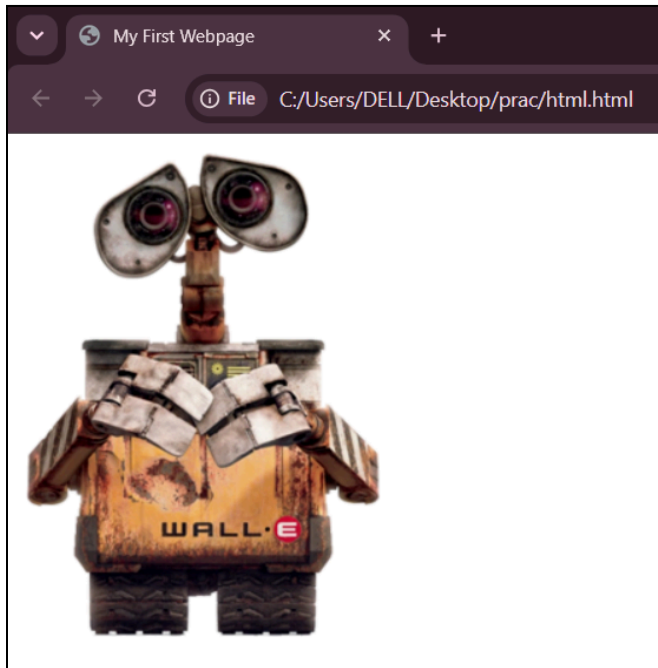
Text
`</p>`

Multiple Attributes:

``

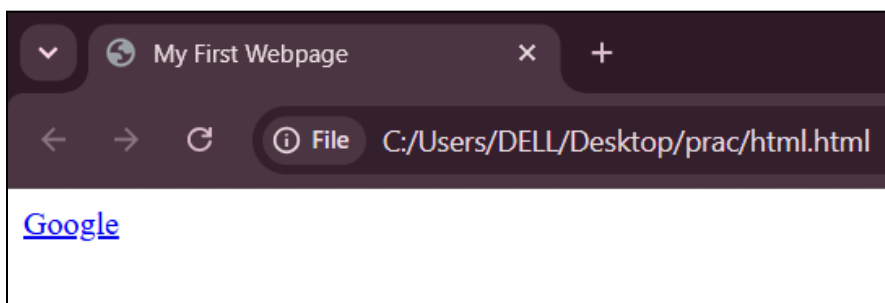
HTML Images

```
<body>  
  
</body>
```



HTML Links

```
<body>  
<a href="https://google.com"> Google </a>  
</body>
```



HTML Tables

```
<body>
```

```
<table border="1">
```

```
<tr>
```

```
  <th>Name</th>
```

```
  <th>ID</th>
```

```
</tr>
```

```
<tr>
```

```
  <td>Arya</td>
```

```
  <td>101</td>
```

```
</tr>
```

```
<tr>
```

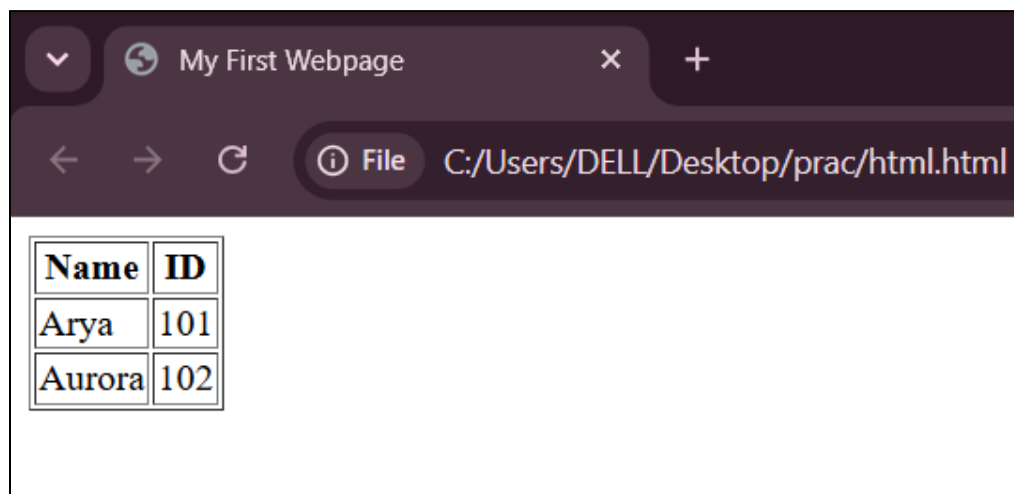
```
  <td>Aurora</td>
```

```
  <td>102</td>
```

```
</tr>
```

```
</table>
```

```
</body>
```



HTML Lists

<body>

Coffee

Tea

Milk

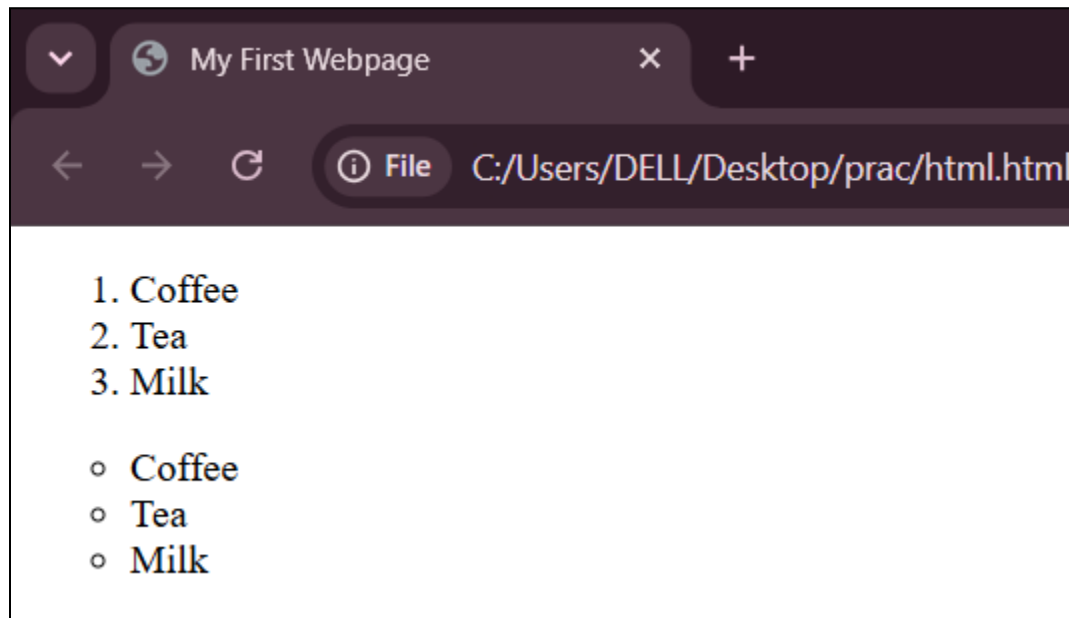
<ul style="list-style-type:circle;">

Coffee

Tea

Milk

</body>



HTML Forms

```
<body>

  <form>

    <div>
      <label for="fname">First Name:</label>
      <input type="text" id="fname" name="fname" placeholder="Tahsin" required>
    </div>

    <br>

    <div>
      <label for="lname">Last Name:</label>
      <input type="text" id="lname" name="lname" placeholder="Tasnim" required>
    </div>

    <br>

    <div>
      <label for="pass">Password:</label>
      <input type="password" id="pass" name="pass" maxlength="12" required>
    </div>

    <br>

    <div>
      <label for="email">Email:</label>
      <input type="email" id="email" name="email"
placeholder="tahsintasnim@gmail.com">
    </div>

    <br>

    <div>
      <label for="bdate">birthdate:</label>
      <input type="date" id="bdate" name="bdate">
    </div>

    <br>
```

```
<div>
  <label for="quantity">quantity:</label>
  <input type="number" id="quantity" name="quantity" min="0" max="99" value="1">
</div>

<br>

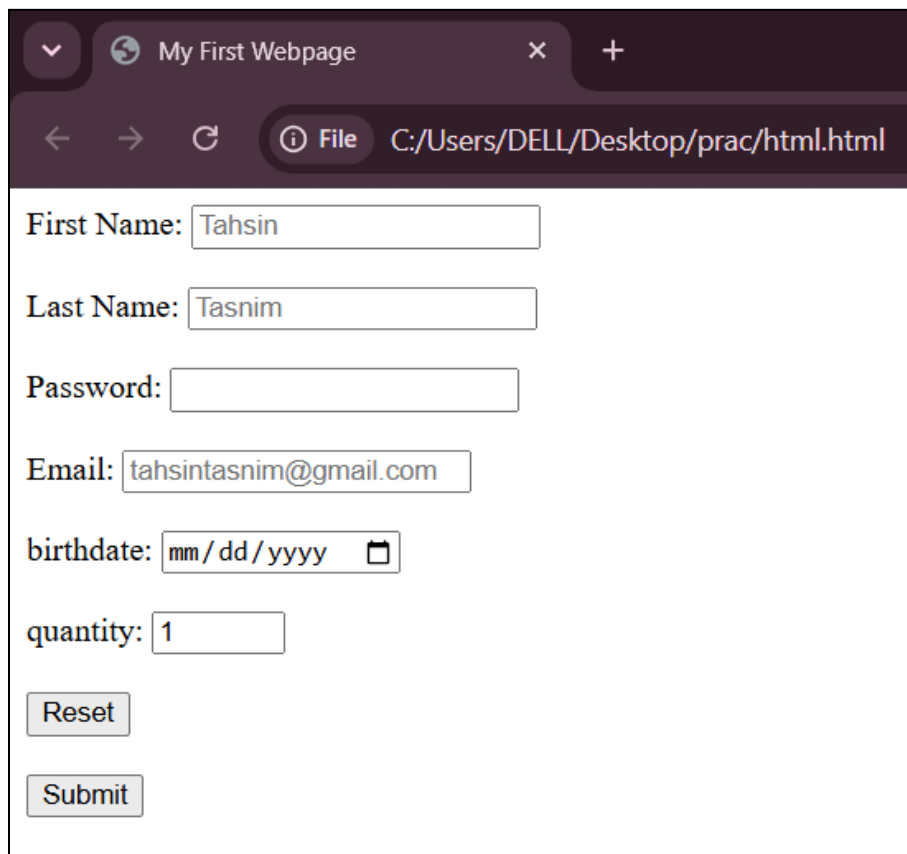
<div>
  <input type="reset">
</div>

<br>

<div>
  <input type="submit">
</div>

</form>

</body>
```



A screenshot of a web browser window titled "My First Webpage". The address bar shows the file path "C:/Users/DELL/Desktop/prac/html.html". The form contains the following elements:

- First Name:
- Last Name:
- Password:
- Email:
- birthdate: with a calendar icon
- quantity:
- Reset button
- Submit button

Head Section

The head contains information about the webpage.

```
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  <title>My Webpage</title>  
</head>
```

meta tag provides metadata:

Character Encoding

```
<meta charset="UTF-8">
```

Supports all languages.

Responsive Design

```
<meta  
name="viewport"  
content="width=device-width, initial-scale=1.0">
```

Makes web pages mobile-friendly.